

Project Report

Optimizing User, Group, and Role Management with Access Control and Workflows

Team Members:

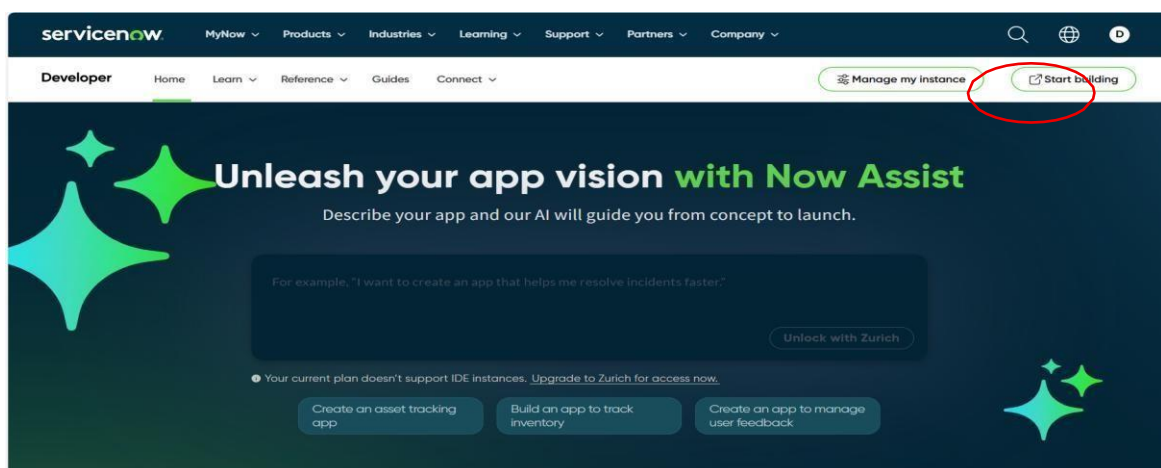
- HARISH S(Team Leader)
- ARAVIND MALAVAN V
- VIGNESH M
- SHIVANAMBI R

Project description:

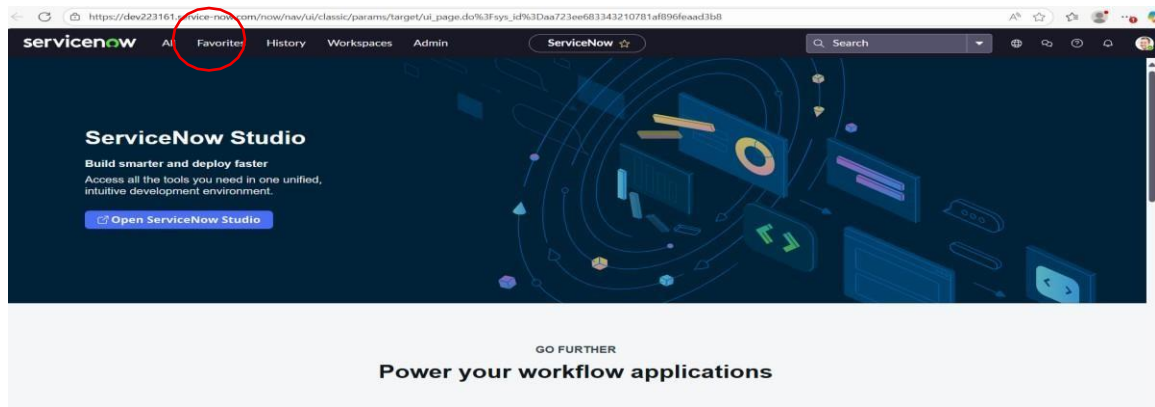
- This project aims to optimize user, group, and role management by implementing automated workflows and access control policies. It ensures secure and efficient access provisioning, reduces manual tasks, and maintains compliance. The system enhances operational efficiency by providing role-based permissions and approval workflows.

Steps for the instance:

1. Sign up for a developer account on the ServiceNow Developer site
“<https://developer.servicenow.com>”.
2. Once logged in, navigate to the "Personal Developer Instance" section.
3. Click on "Request Instance" to create a new ServiceNow instance.
4. Fill out the required information and submit the request.
5. You'll receive an email with the instance details once it's ready.
6. Log in to your ServiceNow instance using the provided credentials.
7. Now you will navigate to the ServiceNow.

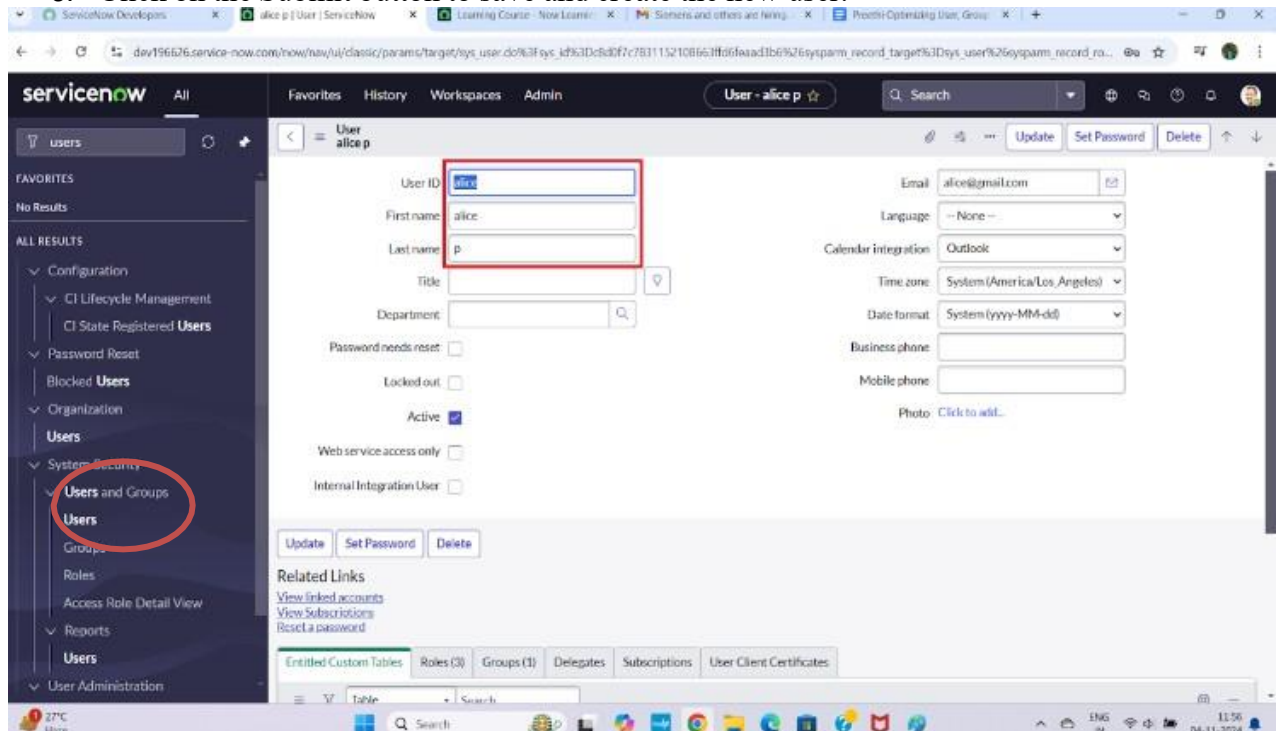


Next:



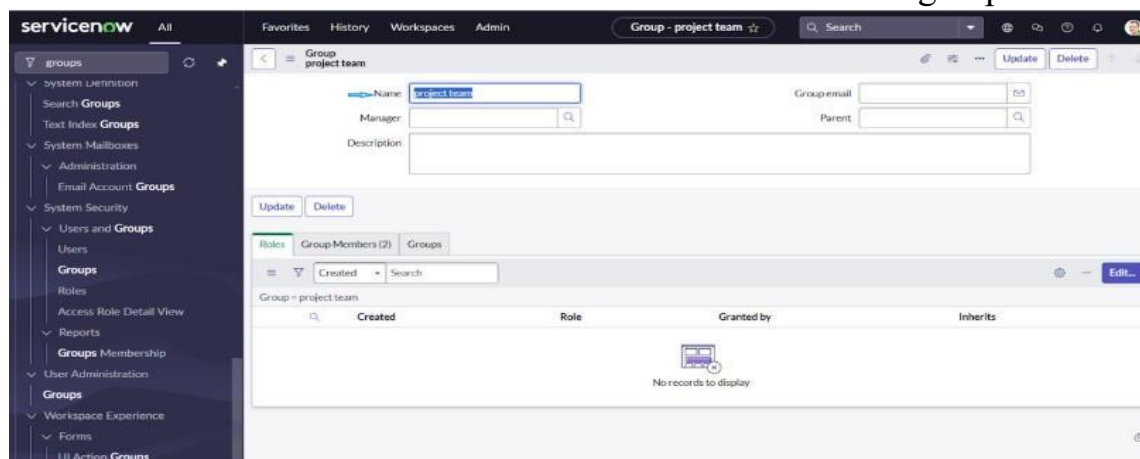
Create Users :

1. Open ServiceNow and log in to your instance.
2. Click on All in the navigation pane and search for Users.
3. Select Users under the System Security module.
4. Click on the New button to create a new user record
5. Fill in the required user details such as Name, User ID, Email, and Password.
6. Click on the Submit button to save and create the new user.



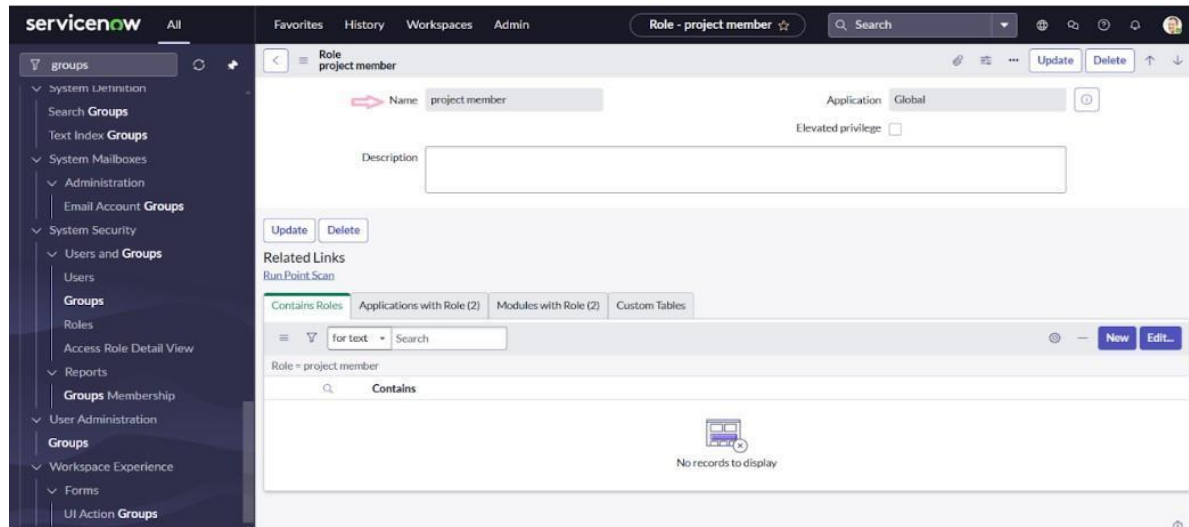
Create Groups :

1. Open ServiceNow and log in to your instance.
2. Click on All in the navigation pane and search for Groups.
3. Select Groups under the System Security module.
4. Click on the New button to create a new group record.
5. Fill in the required group details such as Name, Description, and Manager project team.
6. Click on the Submit button to save and create the new group.



Create Roles :

1. Click on All in the navigation pane and search for Roles.
2. Select Roles under the System Security module.
3. Click on the New button to create a new role record.
4. Fill in the required role details such as Name and Description ☐ project member.
5. Click on the **Submit** button to save and create the new role.

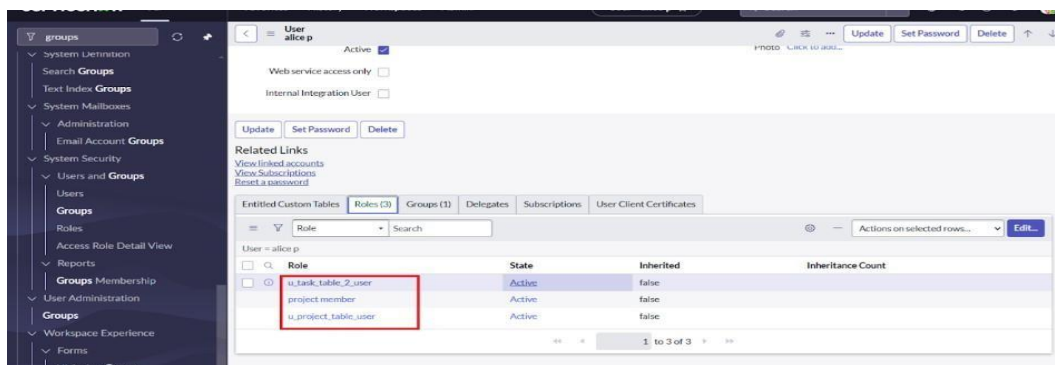


6. Create another role by clicking on the **New** button and filling in the required details such as **Name** and **Description**.
7. Click on the **Submit** button to save and create the additional role.

Assigning users to groups

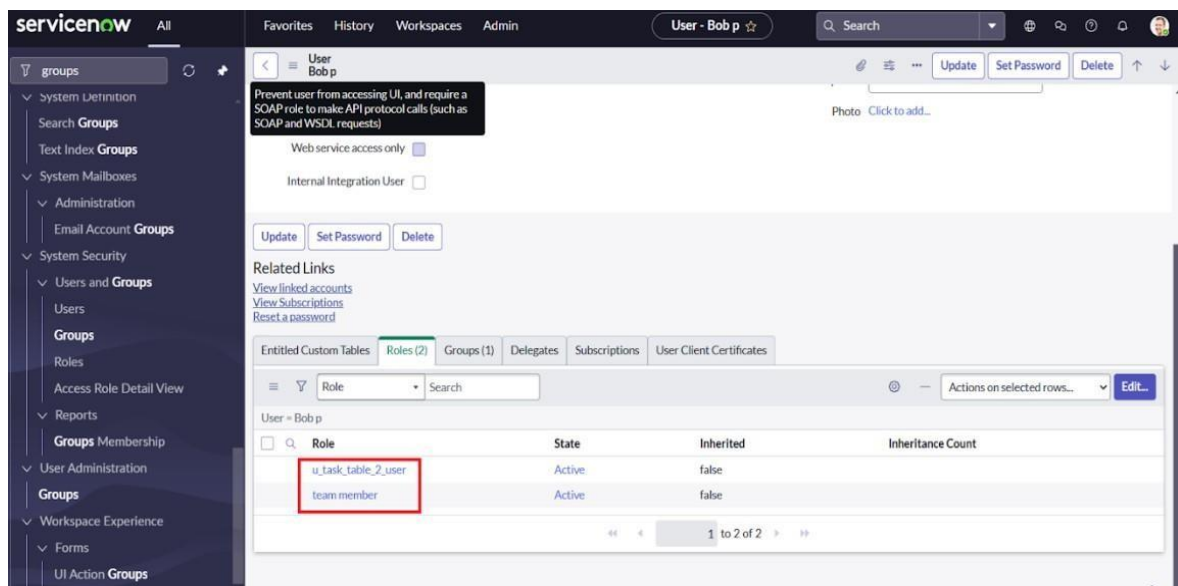
Assign roles to 1st user:

1. Click on All in the navigation pane and search for Users.
2. Select Users under the System Definition module.
3. Open the (Project Manager) user record.
4. Under the Roles related list, click on the Edit button.
5. Select the project_member role and click Save.
6. Again, click Edit and add the u_project_table and u_task_table roles.
7. Click Save, then click Update to finalize and save the changes.



Assign roles to bob user :

1. Click on All in the left navigation pane and search for Users.
2. From the results, select Tables under the System Definition module.
3. Locate and open the Bob P (Team Member) user record from the user list.
4. Scroll down to the Roles section and click on the Edit button to manage user roles.
5. From the available roles, select the team_member role and assign the necessary table roles to Bob.
6. Click on Save and then Update to apply and confirm the changes to Bob's profile.
7. Go to the top-right corner, click on the Profile icon, and choose Impersonate User → Bob to switch to his account.
8. Once impersonated as Bob, navigate through the application to verify that the Task Table is visible and accessible for him.



Application access

Assign table access to application :

1. When you create a new table, ServiceNow automatically generates an application and a module for that table.
2. In the Application Navigator, search for the Project Table application.
3. Click on Edit Module to modify its access settings.
4. Assign the Project Member role to the Project Table application.
5. Next, search for the Task Table2 application and click on Edit Application.
6. Assign both the Project Member and Team Member roles to the Task Table2 application.

servicenow All Favorites History Admin : Application Menu - project table

Search

Application Menu project table

An application menu is a group of modules in the application navigator. Choose the roles that are required to access the application and add or remove modules in the related list below. [More Info](#)

* Title:

Application:

Active: ☒

Restricts access to the specified roles. Otherwise, all users can view the application menu when it is active.

Roles:

Specifies the [menu category](#), which defines the navigation menu style. The default value is Custom Applications.

Category:

The text that appears in a tooltip when a user points to this application menu

Hint:

Description:

Update Delete

Activate Windows
Go to Settings to activate Windows.

ACTIVE ☒

Restricts access to the specified roles. Otherwise, all users can view the application menu when it is active.

Roles:

Specifies the [menu category](#), which defines the navigation menu style. The default value is Custom Applications.

Category:

The text that appears in a tooltip when a user points to this application menu

Hint:

Description:

Update Delete

Activate Windows
Go to Settings to activate Windows.

Modules Order Search

Actions on selected rows... New

Access control list

Creating ACL :

1. Click on All in the left-side navigation pane and type ACL in the search bar to locate access control settings within the platform.
2. From the search results, select Access Control (ACL) under the System Security module to open the list of existing access control rules.
3. Before creating a new ACL, click on Elevate Role at the top-right corner to temporarily gain the security_admin role required for modifying or creating ACLs.

- Once elevated, click on the New button to open the ACL creation form and start defining a new access control rule.
- In the form, carefully fill in the required details such as the Type (record or field), Operation (read, write, delete, etc.), and the Table or Field name for which the ACL will be applied, then add appropriate conditions and roles before saving.

- Scroll down to the **Requires Role** section in the ACL form.
- Double-click to **insert a new row** and add the necessary role permissions.
- Enter the **Task Table** and assign the **Team Member** role to it.
- Click on the **Submit** button to save the ACL configuration.
- Similarly, **create four more ACLs** for the following fields by repeating the same process — defining the table, selecting the appropriate operation, and assigning the required roles before saving each ACL.

Name	Decision Type	Operation	Type	Active	Updated by	Updated
u_leave_request	Allow If	delete	record	true	admin	2024-10-22 02:27:59
u_leave_request	Allow If	create	record	true	admin	2024-10-22 02:27:59
u_task_table	Allow If	read	record	true	admin	2024-10-22 04:21:28
u_task_table	Allow If	write	record	true	admin	2024-10-22 04:20:15

- Click on the **Profile icon** located at the top-right corner of the ServiceNow interface.
- Select the **Impersonate User** option from the dropdown menu.
- Choose the **Bob** user from the list to log in as that user.
- Go to **All** in the application navigator and select **Task Table2** from the application menu bar.
- Verify that the **Comment** and **Status** fields are editable, confirming that Bob has **edit access** to those fields.

task id

task name

status

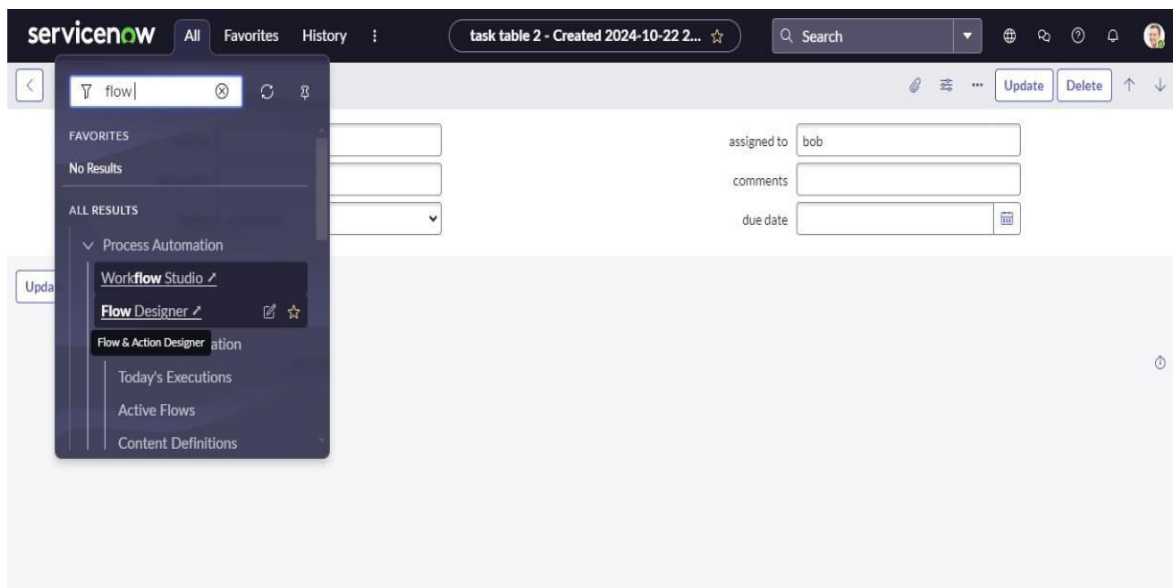
assigned to

comments

due date

Flow

1. Creating a Flow to Assign operations ticket to group :
2. Click on All in the left navigation pane and search for Flow Designer.
3. Select Flow Designer under the Process Automation module to open the flow configuration workspace.
4. Once Flow Designer opens, click on the New button and select Flow from the available options.
5. In the Flow Properties section, enter the Flow Name as “Task Table”.
6. Ensure that the Application is set to Global before proceeding.
7. Finally, click on Build Flow to start designing the workflow for the Task Table.



Workflow Studio

task table
Flow

HomepageOperationsIntegrations

PlaybooksFlowsSubflowsActionsDecision tables

Flows 39
Last refreshed just now

Refresh

Settings

Share

Filter

New ▾
Pick
Playbook
Flow
Subflow
Action
Decision table

☐	Name ▲	Application	Status	Active	Updated
	Benchmark Recommendation Evaluator	Benchmarks Spoke	Published	true	2024-09-27 12:00:00
	Business process approval flow	Global	Published	true	2020-09-27 12:00:00
	Change - Cloud Infrastructure - Authorize	Global	Published	true	2020-11-11 07:08:05
	Change - Emergency - Authorize	Global	Published	true	2020-10-06 05:39:49
	Change - Emergency - Implement	Global	Published	true	2020-09-23 05:06:26
	Change - Emergency - Review	Global	Published	true	2020-10-27 04:18:08
	Change - Normal - Assess	Global	Published	true	2020-10-06 05:37:05
	Change - Normal - Authorize	Global	Published	true	2020-10-06 05:38:35
	Change - Normal - Implement	Global	Published	true	2020-09-23 04:23:59

Late...

New Flow
Flow

Let's get the details for your flow

Flow name *

task table

Description

Describe your flow.

Application *

Global

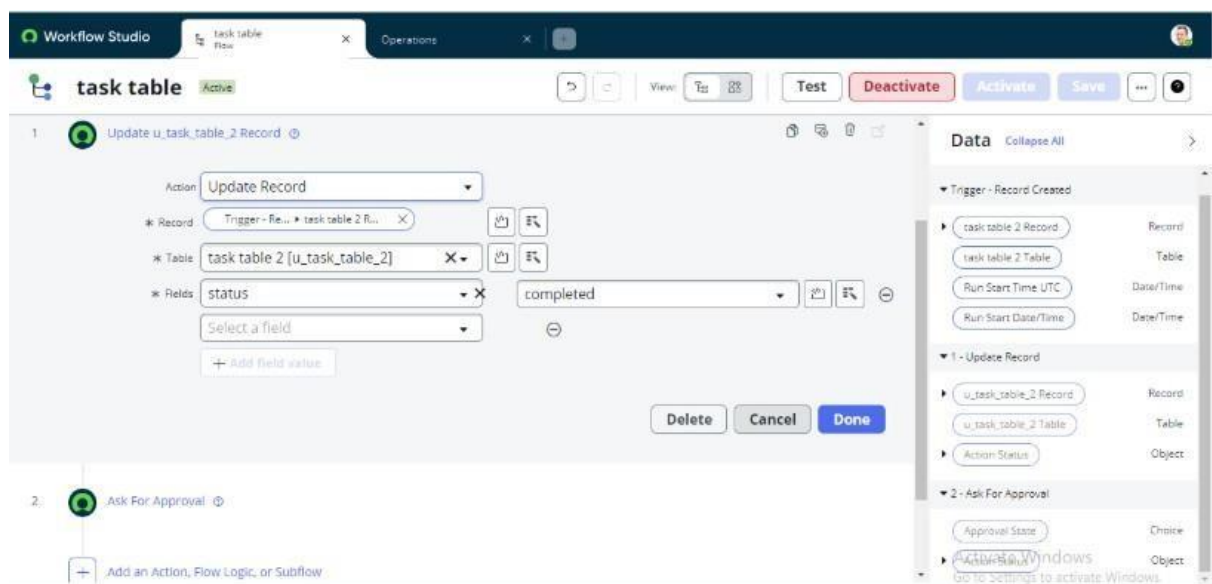
> Show additional properties

Activate Windows
Go to Settings to activate Windows.

CancelBuild flow

Creating Trigger:

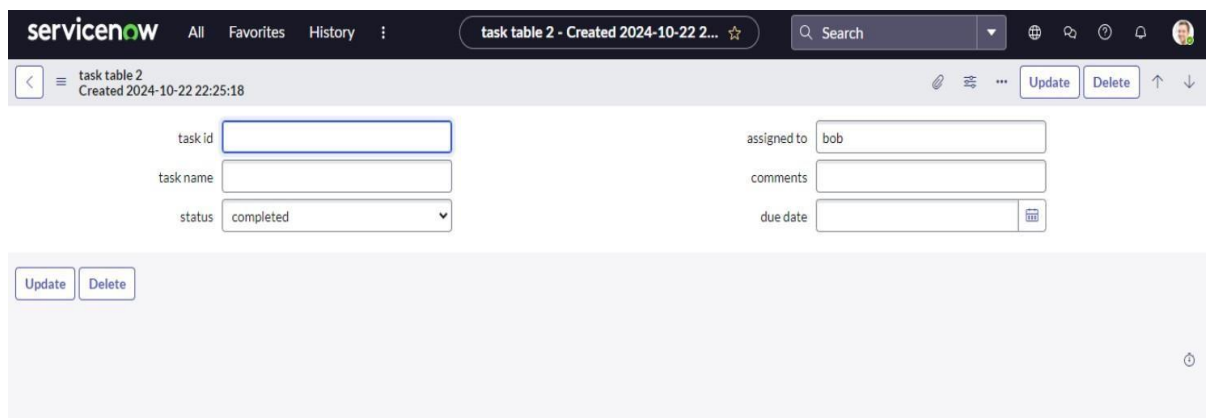
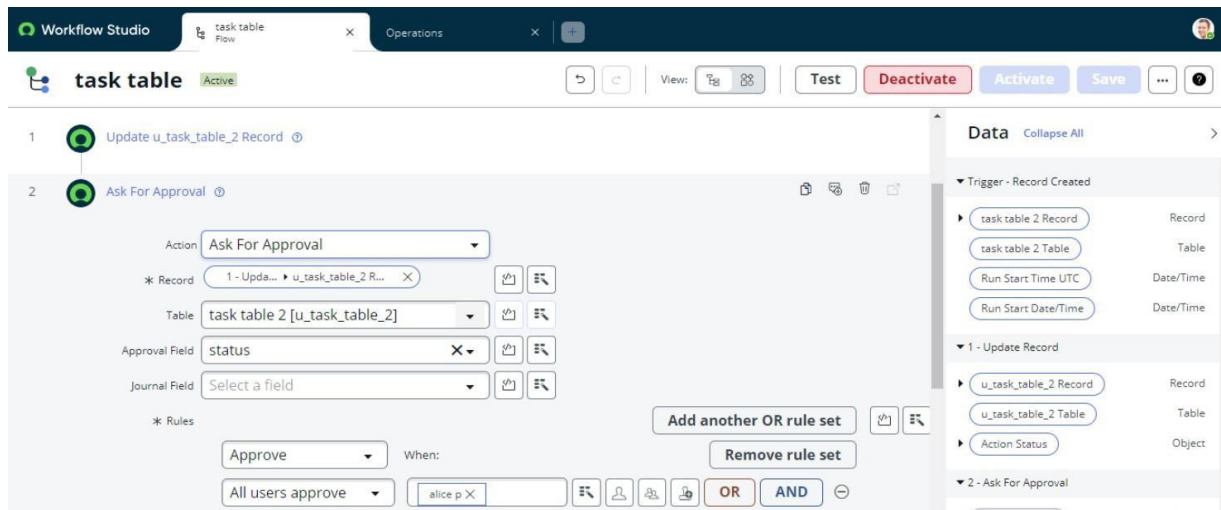
1. Click on **Add a Trigger** to define when the flow should start.
2. In the trigger selection window, search for “**Create Record**” and select it as the trigger type.
3. In the configuration section, set the **Table Name** to “**Task Table**” to specify where the trigger will apply.
4. Define the following **conditions** for the trigger:
 - a. **Field:** Status **Operator:** is **Value:** In Progress
 - b. **Field:** Comments **Operator:** is **Value:** Feedback
 - c. **Field:** Assigned To **Operator:** is **Value:** Bob
5. After entering all the conditions, click on **Done** to save and confirm the trigger configuration.



ACTIONS:

1. Under the **Actions** section of the Flow Designer, click on **Add an Action** to define what happens after the trigger event.
2. In the action selection window, search for “**Ask for Approval**” and select it from the list.
3. In the **Record** field, drag and drop the appropriate fields from the **Data Panel** on the right side to link the record details.
4. The **Table** field will be automatically populated based on the selected record.

5. Set the **Approve Field** to “**Status**” to capture the approval outcome in that field.
6. Assign **Alice P** as the **Approver** who will receive and handle the approval request.
7. Once all details are configured, click on **Done** to save the action settings.



1. Go to the **Application Navigator** in ServiceNow and search for **My Approvals**.
2. Click on **My Approvals** under the **Service Desk** module to view the pending approval requests.
3. You will see that **Alice p** has received an approval request generated by the flow.
4. Right-click on the **Requested** approval record and select **Approve** to complete the approval process.

State	Approver	Comments	Approval for	Created
Approved	alice p		(empty)	2024-10-22 22:26:19
Rejected	Fred Luddy		(empty)	2024-09-01 12:19:33
Requested	Fred Luddy		(empty)	2024-09-01 12:17:03
Requested	Fred Luddy		(empty)	2024-09-01 12:15:44
Requested	Howard Johnson		CHG0000096	2024-09-01 06:15:29
Requested	Ron Kettering		CHG0000096	2024-09-01 06:15:29
Requested	Luke Wilson		CHG0000096	2024-09-01 06:15:29
Requested	Christen Mitchell		CHG0000096	2024-09-01 06:15:29
Requested	Bernard Laboy		CHG0000096	2024-09-01 06:15:29
Requested	Howard Johnson		CHG0000095	2024-09-01 06:15:25
Requested	Ron Kettering		CHG0000095	2024-09-01 06:15:25
Requested	Luke Wilson		CHG0000095	2024-09-01 06:15:25
Requested	Christen Mitchell		CHG0000095	2024-09-01 06:15:25
Requested	Bernard Laboy		CHG0000095	2024-09-01 06:15:25

Result :

- ✓ The project began with the creation of **users, groups, and roles** to establish a structured access control system. Each user was assigned specific roles — **Alice** as the Project Manager and **Bob** as the Team Member — ensuring clear responsibilities. Next, **tables** such as *Project Table* and *Task Table2* were created, and appropriate **roles and permissions** were assigned to control access.
- ✓ **Access Control Lists (ACLs)** were then configured to provide secure data visibility and editing rights. After that, a **Flow Designer** automation was built, where a **trigger** was set to activate whenever a new task record was created under certain conditions.
- ✓ When a task met these conditions, the flow automatically generated an **approval request** for **Alice**. Upon Dharshni's approval, **Bob** could access and update the task details, confirming the workflow's successful execution.
- ✓ Through this process, the project achieved a seamless workflow where **automation, security, and collaboration** worked together — improving efficiency, ensuring proper access control, and enabling smooth project management from start to finish.

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☐	Requested	Bernard Laboy		CHG0000095	2024-09-01 06:15:25

Conclusion :

This scenario demonstrates a well-organized project management process, emphasizing the responsibilities of Alice and Bob within a clearly defined workflow. With Alice managing approvals and Bob handling task execution, the team works together efficiently to achieve project goals. The use of tables helps structure important data, making it easier to monitor projects, tasks, and progress. Overall, this system improves accountability, streamlines communication, and supports the successful delivery of projects.

